

Package Installer

3/5 82 min
3AM

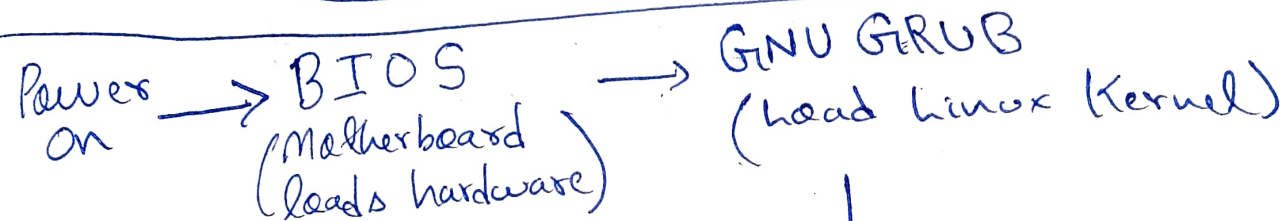
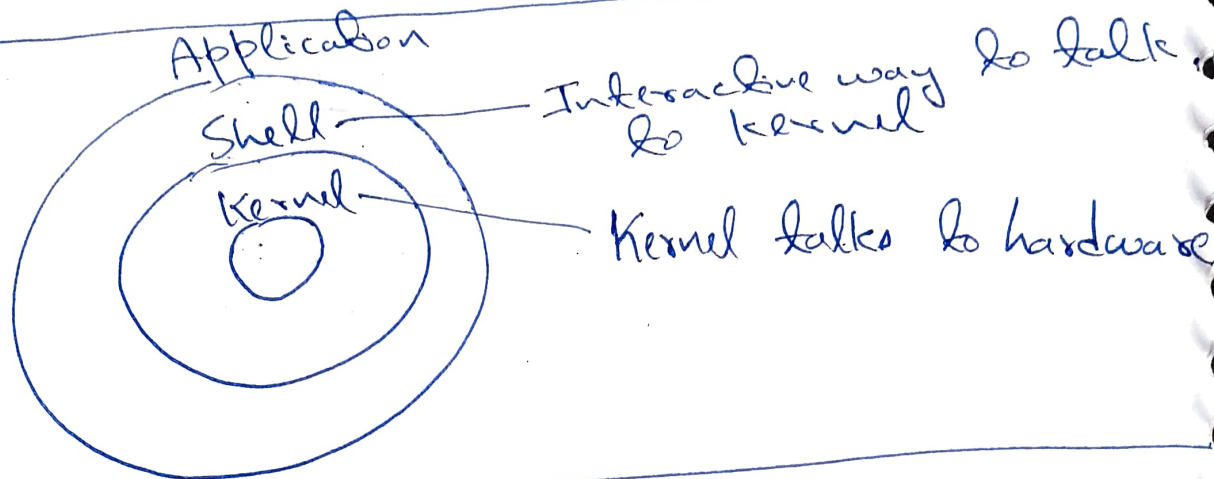
Ubuntu → apt

Redhat → ~~rpm~~ rpm, dnf

CentOS → yum

man apt ← to see manual of command apt

journalctl -u <service> ← to see logs of service



↓
OS loading
↓

init process / systemd
PID → 1

↓
daemon

↓
anything that
runs in
background

systemd की मदद से
किसी भी process की
status देख सकते हैं।
systemctl start nginx
systemctl status nginx

ps → show processes

ps -a → show active processes

top → show processes with memory usage

htop → graphical process memory usage

ps aux | grep ping → $\overline{251}$ $\overline{27}$ ping ~~id~~ process

kill 2514 → process id

a → active running processes

u → processes of user

x → processes not attached to terminal

26th Jan

Linux Process States

B123

Code	State	Description
R	Running/Runnable	Actively executing on CPU, or in the run queue waiting for CPU time
S	Interruptible Sleep	Waiting for an event (I/O, signal, timer); can be waked by signals
D	Uninterruptible Sleep	Waiting on I/O (Usually disk) Cannot be killed or interrupted by signals until the I/O completes
T	Stopped	Process suspended, usually via SIGSTOP, SIGSTP or job control (Ctrl+Z)

Code	State	Description
T	Traced/Stopped	Stopped by a debugger (e.g. during ptrace)
Z	Zombie	Process has terminated but its parent hasn't yet reaped its exit status via wait()
X	Dead	Process is being terminated (rarely - very transient state)

Additional Flags (appended to the state)
 These modify the base state and appear as extra characters

L → high priority (not nice to other processes)

N → low priority (Nice)

L → Pages locked into memory

S → session leader

R → multi-threaded

+ → in the foreground process group

Ex Ss → interruptible sleep, session leader

Ex R+ → running, in foreground

★ D state is the dangerous one

A process stuck in D usually means a hung disk, NFS mount, or failing storage device. We can't kill -9 it, we need to fix the underlying I/O issue or wait it out.

★ Z (zombie) processes aren't using resources themselves (memory/CPU already freed) but consume a PID slot. Too many zombies usually points to a parent process that isn't calling wait() properly → check the parent, not the zombie

Ex `ps -eo pid,ppid,stat,cmd | grep D`

Ex `ps aux | awk '$8 ~ /Z/ {print}'`

top shows ~~the~~ live count of processes by state at the top

Tasks: X total, Y running, Z sleeping
